Lab Report

BA-MECH-24-4C

Control Engineering II

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Chapter 1

Introduction

1.1 General Description

This report documents the modelling, identification and position control of a DC motor that drives a pointer via a gearbox with belt drive (Box 1 of the laboratory test benches). The controlled variable is the angular position of the pointer, measured on the output shaft by an optical quadrature encoder. The motor is supplied by a PWM-controlled motor driver, and all signals are processed by a data acquisition card with MATLAB/Simulink and Simulink Desktop Real-Time [1].

1.2 Motivation

DC motors are widely used electro-mechanical actuators, for example in automotive applications, medical devices and industrial automation [1]. An exact positioning of the driven load requires a closed-loop controller, whose design relies on a sufficiently accurate model of the plant. Since not all parameters of a real drive train are known, the model is derived both analytically and from measured data.

1.3 Learning Goal

The aim of the laboratory is to identify an electro-mechanical system from its measured step response and to design a suitable controller [1]. The step response is approximated by a second-order lag element, a PID controller is designed with the Ziegler–Nichols method, and the closed loop is simulated, implemented on the real test bench and evaluated.

Chapter 2

Procedure and Method

2.1 Experimental Setup

The equipment is listed in Table 2.1; the labelled test bench is shown in Figure A.1 in Appendix A.

Table 2.1: List of equipment used for the experiment [1]

Component	Description	Type
Test bench	gearbox with belt drive, Box 1, $N = 97.2$	–
DC motor	brushed DC motor, nominal voltage $12\,\mathrm{V}$	Faulhaber 3257G012CR [2]
Motor driver	PWM-controlled motor driver	Cytron MD13S [3]
Encoder	quadrature encoder, 500 counts per revolution	Broadcom HEDS-5500 A11 [4]
DAQ card	data acquisition card	NI PCIe-6321 [5]
I/O module	BNC connector block	NI BNC-2090a
Software	identification, simulation, real-time control	MATLAB/Simulink R2020b, R2024b

2.2 Analytical Modelling of the DC Motor

The unloaded DC motor is described by a white-box model based on the schematic representation in Figure A.2 in Appendix A. In this report, the motor quantities are written without the index “mot”, U_{in} is denoted by u, U_{emf} by u_{emf}, and the rotation n_{out} is described by the angular velocity ω. All symbols are explained in the List of Symbols. Kirchhoff’s voltage law for the armature circuit with the back electromotive force $u_{\mathrm{emf}} = k_{\mathrm{E}}\,\omega$ yields Equation (2.1), the balance of torques at the rotor yields Equation (2.2).

$$u(t) = R\,i(t) + L\,\frac{\mathrm{d}i(t)}{\mathrm{d}t} + k_{\mathrm{E}}\,\omega(t) \tag{2.1}$$

$$J\,\frac{\mathrm{d}\omega(t)}{\mathrm{d}t} = k_\mathrm{M}\,i(t) - b\,\omega(t) \tag{2.2}$$

With $\omega = \mathrm{d}\theta/\mathrm{d}t$ for the angular position θ, the Laplace transformation with zero initial conditions gives

$$U(s) = (R + Ls)\,I(s) + k_\mathrm{E}\,s\,\Theta(s), \qquad Js^2\,\Theta(s) = k_\mathrm{M}\,I(s) - b\,s\,\Theta(s),$$

where capital letters denote Laplace transforms and s is the Laplace variable. Solving the mechanical equation for $I(s) = (Js^2 + bs)\,\Theta(s)/k_\mathrm{M}$ and substituting it into the electrical equation yields

$$U(s) = \left[\frac{(R + Ls)(Js^2 + bs)}{k_\mathrm{M}} + k_\mathrm{E}\,s\right]\Theta(s) = \frac{s\left[LJs^2 + (RJ + Lb)\,s + (Rb + k_\mathrm{E}k_\mathrm{M})\right]}{k_\mathrm{M}}\,\Theta(s).$$

The transfer function from the armature voltage to the angular position in Equation (2.3) is therefore a third-order system with an integrator. With $\Omega(s) = s\,\Theta(s)$, the transfer function to the angular velocity in Equation (2.4) is a second-order lag element (PT2), which justifies the PT2 approach in Section 2.3.

$$G_\mathrm{p1}(s) = \frac{\Theta(s)}{U(s)} = \frac{k_\mathrm{M}}{s\left[LJs^2 + (RJ + Lb)\,s + (Rb + k_\mathrm{E}k_\mathrm{M})\right]} \tag{2.3}$$

$$G_\mathrm{p2}(s) = \frac{\Omega(s)}{U(s)} = s\,G_\mathrm{p1}(s) = \frac{k_\mathrm{M}}{LJs^2 + (RJ + Lb)\,s + (Rb + k_\mathrm{E}k_\mathrm{M})} \tag{2.4}$$

With the state variables $\mathbf{x} = \begin{bmatrix}\theta & \omega & i\end{bmatrix}^\mathrm{T}$, the input u and the output θ, rearranging Equations (2.1) and (2.2) for the derivatives gives the state-space model

$$\dot{\mathbf{x}} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & -\dfrac{b}{J} & \dfrac{k_\mathrm{M}}{J} \\ 0 & -\dfrac{k_\mathrm{E}}{L} & -\dfrac{R}{L} \end{bmatrix}}_{\mathbf{A}} \mathbf{x} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ \dfrac{1}{L} \end{bmatrix}}_{\mathbf{b}} u, \qquad \theta = \underbrace{\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}}_{\mathbf{c}^\mathrm{T}} \mathbf{x}.$$

2.3 System Identification

The step response is recorded for an input step of $u_0 = 12\,\mathrm{V}$ at $t = 0\,\mathrm{s}$ with a sample time of $0.1\,\mathrm{ms}$ [1]. The encoder provides the rotational speed n of the output shaft in min^{-1}, which is converted into $\omega = 2\pi n/60\,\mathrm{s\,min}^{-1}$; ω thus refers to the output shaft, which is slower than the motor shaft by the factor N. The plant is approximated by an overdamped PT2 element according to Equa-

tion (2.5) with the static gain K in $\mathrm{rad\,s^{-1}\,V^{-1}}$ and the time constants τ_1, τ_2 in s.

$$G_{\mathrm{PT2}}(s) = \frac{\Omega(s)}{U(s)} = \frac{K}{(\tau_1 s + 1)(\tau_2 s + 1)} \tag{2.5}$$

K, τ_1 and τ_2 are determined by a least-squares fit of the step response

$$\omega(t) = u_0 K \left[1 - \frac{\tau_1 \,\mathrm{e}^{-t/\tau_1} - \tau_2 \,\mathrm{e}^{-t/\tau_2}}{\tau_1 - \tau_2}\right]$$

to the measurement with the MATLAB function `lsqcurvefit`. The preparatory identification is given in Appendix B. Integrating the angular velocity gives the plant for the position control in Equation (2.6) [1].

$$G_{\mathrm{p}}(s) = \frac{\Theta(s)}{U(s)} = \frac{1}{s}\, G_{\mathrm{PT2}}(s) = \frac{K}{s\,(\tau_1 s + 1)(\tau_2 s + 1)} \tag{2.6}$$

2.4 Controller Design and Implementation

A PID controller according to Equation (2.7) with the proportional gain K_{P} in $\mathrm{V\,rad^{-1}}$, the integral and derivative times T_{I}, T_{D} in s, $K_{\mathrm{I}} = K_{\mathrm{P}}/T_{\mathrm{I}}$ and $K_{\mathrm{D}} = K_{\mathrm{P}}\,T_{\mathrm{D}}$ is designed with the closed-loop method of Ziegler and Nichols [6].

$$G_{\mathrm{c}}(s) = K_{\mathrm{P}}\left(1 + \frac{1}{T_{\mathrm{I}}\,s} + T_{\mathrm{D}}\,s\right) = K_{\mathrm{P}} + \frac{K_{\mathrm{I}}}{s} + K_{\mathrm{D}}\,s \tag{2.7}$$

The critical gain K_{cr}, at which the closed loop of $G_{\mathrm{p}}(s)$ with a pure P controller reaches the stability limit, is determined from the root locus with the MATLAB function `rlocus`, and the period P_{cr} of the sustained oscillation from the simulated output signal [1]. The tuning rules $K_{\mathrm{P}} = 0.6\,K_{\mathrm{cr}}$, $T_{\mathrm{I}} = 0.5\,P_{\mathrm{cr}}$ and $T_{\mathrm{D}} = 0.125\,P_{\mathrm{cr}}$ then give the controller parameters [6]. The closed loop is simulated in Simulink, with the reference converted from degrees into radians before the controller and the output converted back [1].

The controller is implemented in Simulink Desktop Real-Time with a sample time of $0.1\,\mathrm{ms}$, in the parallel form of Equation (2.7) with a first-order derivative filter (coefficient $100\,\mathrm{s^{-1}}$). The control value is limited to $\pm 12\,\mathrm{V}$, and integrator windup is prevented by clamping. For each run, a reference step of $45°$ is recorded for $1\,\mathrm{s}$, while K_{P}, K_{I} and K_{D} are varied one after another. A manual disturbance and a sinusoidal reference are also tested.

Chapter 3

Results

3.1 Identified Transfer Function

The least-squares fit yields the parameters $K = 0.4334\,\mathrm{rad\,s^{-1}\,V^{-1}}$, $\tau_1 = 0.0076\,\mathrm{s}$ and $\tau_2 = 0.1030\,\mathrm{s}$. Measurement and identified model are compared in Figure 3.1. As an example of the conversion of the measured data, the last recorded sample with $n = 50\,\mathrm{min^{-1}}$ corresponds to

$$\omega = \frac{2\pi\, n}{60\,\mathrm{s\,min^{-1}}} = \frac{2\pi \cdot 50\,\mathrm{min^{-1}}}{60\,\mathrm{s\,min^{-1}}} = 5.24\,\mathrm{rad\,s^{-1}}.$$

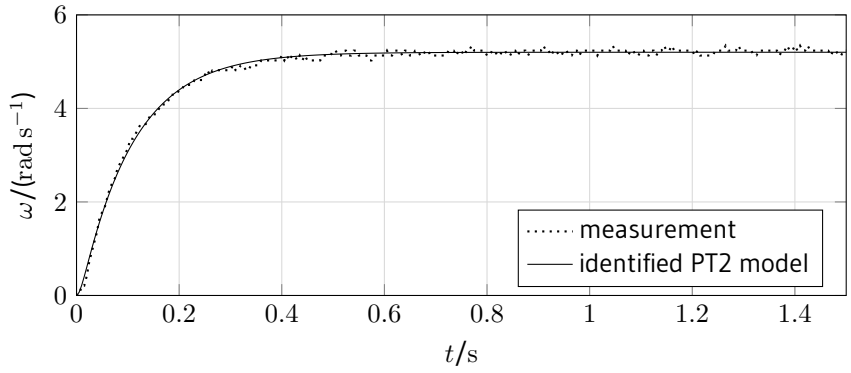


Figure 3.1: Measured angular velocity of the output shaft (dotted) and step response of the identified PT2 model (solid) for an input step of $12\,\mathrm{V}$

With Equation (2.6), the plant for the position control is

$$G_\mathrm{p}(s) = \frac{0.4334\,\mathrm{rad\,s^{-1}\,V^{-1}}}{s\,(0.0076\,\mathrm{s}\cdot s + 1)(0.1030\,\mathrm{s}\cdot s + 1)}.$$

3.2 Controller Parameters and Simulation

From the root locus of $G_{\mathrm{p}}(s)$, a critical gain of $K_{\mathrm{cr}} = 325\,\mathrm{V\,rad^{-1}}$ and a period of $P_{\mathrm{cr}} = 0.1757\,\mathrm{s}$ are obtained. The tuning rules give $K_{\mathrm{P}} = 0.6 \cdot 325\,\mathrm{V\,rad^{-1}} = 195\,\mathrm{V\,rad^{-1}}$, $T_{\mathrm{I}} = 0.5 \cdot 0.1757\,\mathrm{s} = 0.0879\,\mathrm{s}$ and $T_{\mathrm{D}} = 0.125 \cdot 0.1757\,\mathrm{s} = 0.0220\,\mathrm{s}$, and thus

$$K_{\mathrm{I}} = \frac{K_{\mathrm{P}}}{T_{\mathrm{I}}} = \frac{195\,\mathrm{V\,rad^{-1}}}{0.0879\,\mathrm{s}} = 2219.7\,\mathrm{V\,rad^{-1}\,s^{-1}}$$

and $K_{\mathrm{D}} = K_{\mathrm{P}}\,T_{\mathrm{D}} = 195\,\mathrm{V\,rad^{-1}} \cdot 0.0220\,\mathrm{s} = 4.28\,\mathrm{V\,s\,rad^{-1}}$. The simulated step response of the closed loop is shown in Figure 3.2.

(Placeholder – MATLAB: `PID_simulation.slx`, Scope of the closed loop)

Figure 3.2: Simulated angular position of the closed loop with the PID controller

3.3 Real-Time Control

Table 3.1 lists the controller gains of all runs with a reference step of $45°$ together with the observations noted during the laboratory. The measured step responses are shown in Figures C.1 to C.8 in Appendix C.

Table 3.1: Controller gains and observations of the real-time runs for a reference step of 45°

Run	$K_\mathrm{P}/(\mathrm{V\,rad^{-1}})$	$K_\mathrm{I}/(\mathrm{V\,rad^{-1}\,s^{-1}})$	$K_\mathrm{D}/(\mathrm{V\,s\,rad^{-1}})$	Observation
1	195	22.2	4.28	K_I adjusted
2	195	2219.7	4.28	calculated (Ziegler–Nichols)
3	1	22.2	4.28	K_P reduced
4	20	22.2	4.28	K_P reduced
5	195	22.2	8.00	K_D increased
6	195	22.2	1.00	K_D reduced, overshoot
7	300	22.2	4.28	K_P increased
8	300	22.2	6.00	K_P and K_D increased

With the calculated Ziegler–Nichols parameters (run 2), the position overshoots the reference before it settles (Figure C.2). With K_I reduced by a factor of 100 (run 1), the reference is reached without visible overshoot (Figure C.1); these parameters are the starting point for the further variations. With the strongly reduced proportional gains of runs 3 and 4, the position rises slowly, is still moving at the end of the recording and exceeds the reference. Increasing K_D (run 5) leads to a response without overshoot, whereas reducing K_D (run 6) causes an overshoot. The step responses with increased K_P (run 7) and additionally increased K_D (run 8) are shown in Figures C.7 and C.8. In all runs, the recorded control value reaches the saturation limit of $12\,\mathrm{V}$ directly after the reference step.

With the gains of run 1, a disturbance is applied manually to the pointer (Figure D.1 in Appendix D): the pointer is deflected by approximately 2° and is returned to the reference by the controller. For the tracking of a sinusoidal reference (Table D.1 and Figures D.2 to D.7 in Appendix D), the response with $K_\mathrm{P} = 195\,\mathrm{V\,rad^{-1}}$ is observed to be slightly slow. Increasing K_P to $300\,\mathrm{V\,rad^{-1}}$ halves the control error, and $600\,\mathrm{V\,rad^{-1}}$ reduces it further, while at $800\,\mathrm{V\,rad^{-1}}$ and $1200\,\mathrm{V\,rad^{-1}}$ the control value becomes strongly noisy.

Chapter 4

Discussion and Interpretation

4.1 Comparison of Measurement, Model and Simulation

The identified PT2 model reproduces the measured step response of the angular velocity well (Figure 3.1): both the rise and the stationary value are matched, and the remaining deviations are mainly caused by the noise of the measured signal. This noise results from the calculation of the rotational speed from the position change of the incremental encoder between two samples [1], so that the speed is only available in discrete steps. The overdamped PT2 structure therefore describes the dynamic behaviour of the drive sufficiently for the controller design, which is consistent with the analytical model in Equation (2.4). The same approach was already successful for the preparatory data, where an additional dead time was required to describe the delayed reaction of the system (Appendix B).

A comparison with the analytical model shows clear differences. Evaluating Equation (2.4) with the datasheet values of the motor [2] yields a static gain of $52.36\,\mathrm{rad\,s^{-1}\,V^{-1}}$ and time constants of $4.5\,\mathrm{ms}$ and $0.18\,\mathrm{ms}$. The static gain refers to the motor shaft, whereas the identified gain refers to the output shaft; both are therefore only comparable when the transmission ratio N is taken into account, which brings them into the same order of magnitude. The time constants, however, differ substantially: the dominant identified time constant $\tau_2 = 0.1030\,\mathrm{s}$ is more than an order of magnitude larger than the analytical one. The main reason is the assumption of an unloaded motor in the analytical model. The real test bench additionally contains the gearbox, the belt drive and the pointer, whose inertia and friction slow down the system considerably. Furthermore, the motor driver and the real friction behaviour are not part of the analytical model. The analytical model is thus suitable to justify the model structure, while the parameters for the controller design have to be identified experimentally.

4.2 Controller Performance

The controller designed with the Ziegler–Nichols method stabilises the position control loop, but the step response with the calculated parameters (run 2) shows an overshoot (Figure C.2). This is expected, since the tuning rules of Ziegler and Nichols aim at a fast response with a decay ratio of about one quarter rather than at a response without overshoot [6]. The Ziegler–Nichols parameters are therefore a starting point that has to be adjusted on the real system.

The most effective adjustment is the reduction of the integral gain by a factor of 100 (run 1), after which no overshoot is visible (Figure C.1). This can be explained by the structure of the plant: $G_{\mathrm{p}}(s)$ already contains an integrator, so that a constant reference is reached without remaining control error even without integral action. A large integral gain mainly accumulates the control error during the rise and thus causes the overshoot. A small integral gain is nevertheless useful, since it compensates constant disturbances such as friction torques. This is confirmed by the disturbance test (Figure D.1), in which the manually deflected pointer is returned to the reference.

The variation of the gains shows the expected tendencies. A strongly reduced proportional gain (runs 3 and 4) leads to a slow response that has not settled at the end of the recording (Figures C.3 and C.4). Since the proportional part provides only a small control value in this case, the response is presumably dominated by the derivative part directly after the step and by the slowly growing integral part afterwards, which drives the position beyond the reference. The derivative gain acts as damping: increasing it (run 5) suppresses the overshoot, reducing it (run 6) causes an overshoot. An increased proportional gain (runs 7 and 8) is possible, and in combination with a larger derivative gain (run 8), a well-damped response is obtained.

In all runs, the control value reaches the saturation limit of $12\,\mathrm{V}$ directly after the reference step. During this phase, the system is driven with the maximum voltage, so that the rise of the position is limited by the actuator rather than by the controller gains. This explains why the initial rise of the step responses with different gains is very similar (Appendix C), and it shows that the linear design does not fully describe the real behaviour. The clamping anti-windup prevents the integrator from accumulating during the saturation.

The tracking of the sinusoidal reference illustrates the trade-off in the choice of the proportional gain

(Table D.1). A higher gain reduces the control error, but it also amplifies the measurement noise of the encoder signal, which results in a strongly fluctuating control value at $800\,\mathrm{V\,rad^{-1}}$ and above (Figures D.6 and D.7). Such a control value increases the load on the motor and the motor driver; a proportional gain of about $600\,\mathrm{V\,rad^{-1}}$ is therefore a reasonable compromise for this reference signal.

The results are subject to several limitations. Each parameter set is measured only once, so that the reproducibility is not verified, and the recording time of $1\,\mathrm{s}$ is too short to assess the settling of runs 3 and 4. The manual disturbance is not reproducible in magnitude and timing. In addition, the plant model is identified from a single step amplitude of $12\,\mathrm{V}$, so that nonlinear effects such as static friction are not captured.

4.3 Conclusion

The DC motor test bench is successfully modelled, identified and controlled. The analytical model confirms the PT2 structure of the drive, while the parameters of the real system, including gearbox and belt drive, are obtained by a least-squares fit of the measured step response. The Ziegler–Nichols method yields a stable PID position controller, which, however, causes an overshoot on the real system. Reducing the integral gain removes the overshoot, and a combination of an increased proportional and derivative gain (run 8) gives a fast and well-damped response. The experiments further show that the actuator saturation dominates the behaviour after large reference steps and that the achievable proportional gain is limited by the measurement noise. For further investigations, repeated measurements, longer recording times and an identification at several step amplitudes, including the friction behaviour, are recommended.

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List of Symbols

Symbol	Description	Unit
$\mathbf{A}$	system matrix of the state-space model	–
b	viscous friction coefficient	$\mathrm{N\,m\,s\,rad^{-1}}$
$\mathbf{b}$	input vector of the state-space model	–
$\mathbf{c}$	output vector of the state-space model	–
e	control error (reference minus actual position)	$^\circ$
G	transfer function	–
i	armature current	A
J	moment of inertia of the rotor	$\mathrm{kg\,m^2}$
k	motor constant (k_M: torque constant, k_E: back-EMF constant)	$\mathrm{N\,m\,A^{-1}}$, $\mathrm{V\,s\,rad^{-1}}$
K	gain (K: static gain of the plant; K_P, K_I, K_D: controller gains; K_cr: critical gain)	–
L	armature inductance	H
n	rotational speed of the output shaft	$\mathrm{min^{-1}}$
N	transmission ratio of the gearbox	1
P	period of oscillation	s
R	armature resistance	Ω
s	Laplace variable	$\mathrm{s^{-1}}$
t	time	s
T	time constant (T_I: integral time, T_D: derivative time, T_t: dead time)	s
u	voltage (u: armature voltage; u_in, u_out: input and output voltage of the preparatory system)	V
$\mathbf{x}$	state vector	–
θ	angular position	rad
τ	time constant of the PT2 model	s
ω	angular velocity	$\mathrm{rad\,s^{-1}}$

Chapter A

Test Bench

The test bench with its labelled components is shown in Figure A.1, the schematic representation used for the analytical model in Section 2.2 in Figure A.2.

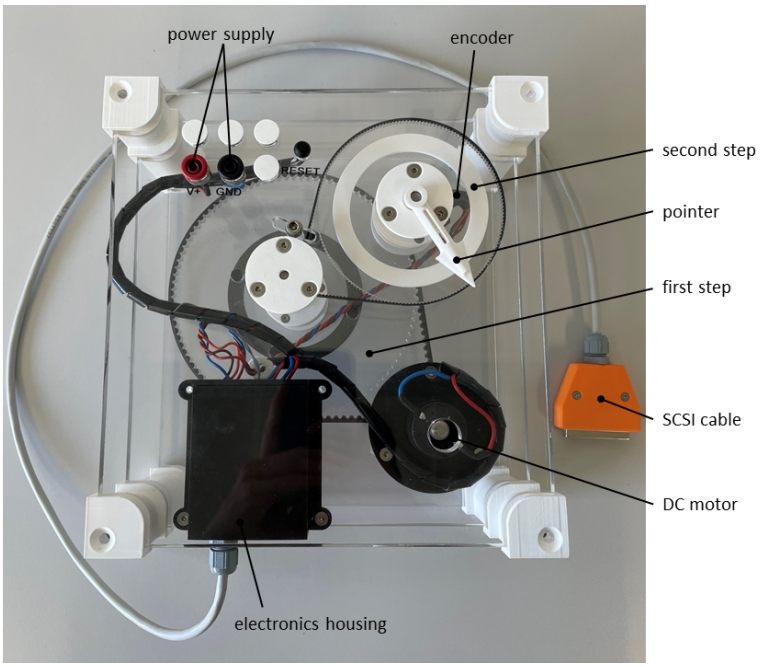


Figure A.1: Mechanical belt transmission system driven by a DC motor with labelled components [1]

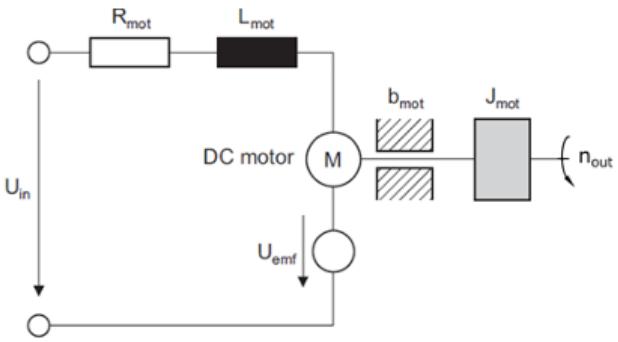


Figure A.2: Schematic representation of the electro-mechanical system of the DC motor [1]

Chapter B

Preparatory System Identification

As preparatory work, the step response of an arbitrary mechatronic system is identified. The measurement data are provided in the file `measurement_data.mat` [1]. Since the output reacts only after a delay, the PT2 element of Equation (2.5) is extended by a dead time T_t according to Equation (B.1).

$$G(s) = \frac{K\,\mathrm{e}^{-T_\mathrm{t}s}}{(\tau_1 s + 1)(\tau_2 s + 1)} \tag{B.1}$$

The measured output voltage is normalised to the step amplitude of $5.24\,\mathrm{V}$, and the parameters are determined by a least-squares fit with the MATLAB function `lsqcurvefit`. The fit yields $K = 4.9344$, $\tau_1 = 0.0931\,\mathrm{s}$, $\tau_2 = 0.0253\,\mathrm{s}$ and $T_\mathrm{t} = 0.3140\,\mathrm{s}$. The raw measurement data and the step response of the identified model are shown in Figure B.1.

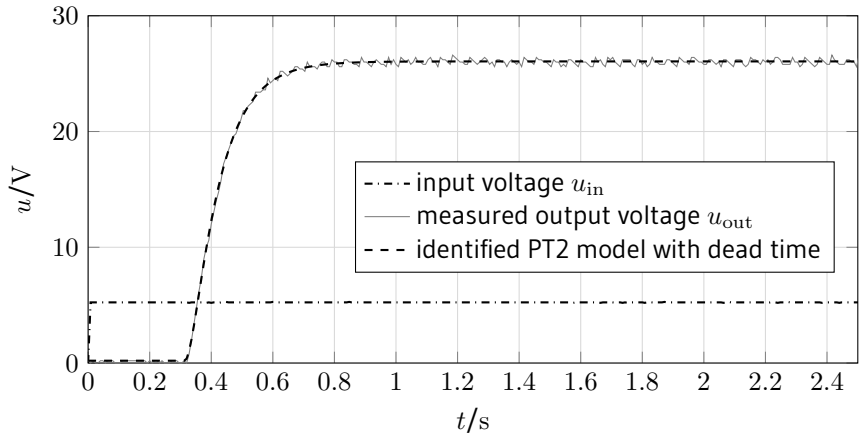


Figure B.1: Raw measurement data of the step response provided in [1] (input dash-dotted, output grey) and step response of the identified PT2 model with dead time (dashed)

Chapter C

Measured Step Responses

The measured step responses of all runs in Table 3.1 are shown in Figures C.1 to C.8.

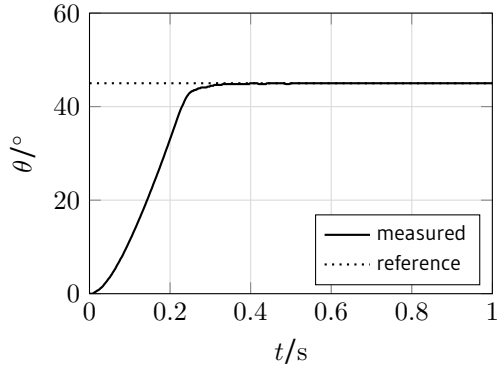


Figure C.1: Measured step response of run 1 (controller gains see Table 3.1)

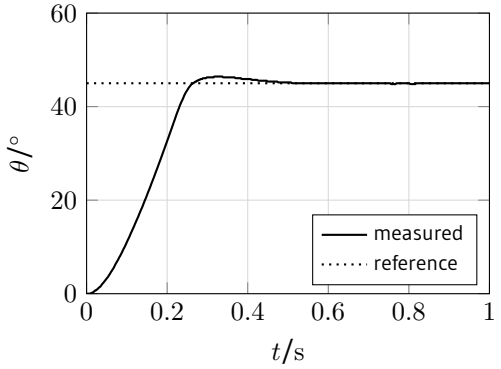


Figure C.2: Measured step response of run 2 (controller gains see Table 3.1)

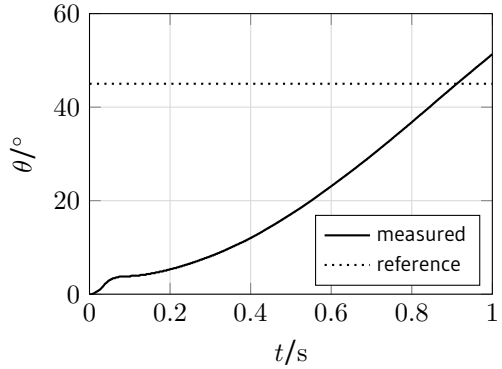


Figure C.3: Measured step response of run 3 (controller gains see Table 3.1)

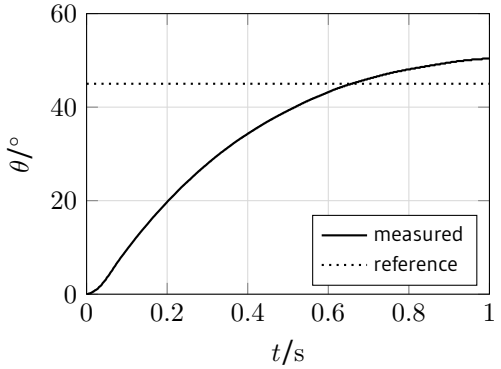


Figure C.4: Measured step response of run 4 (controller gains see Table 3.1)

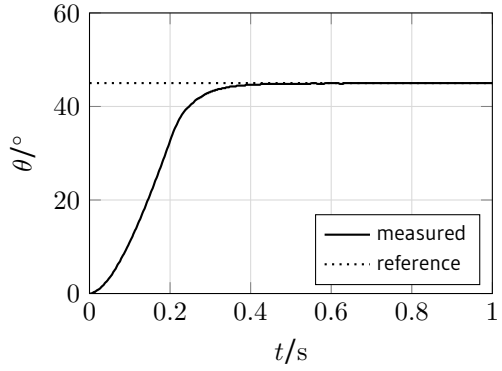


Figure C.5: Measured step response of run 5 (controller gains see Table 3.1)

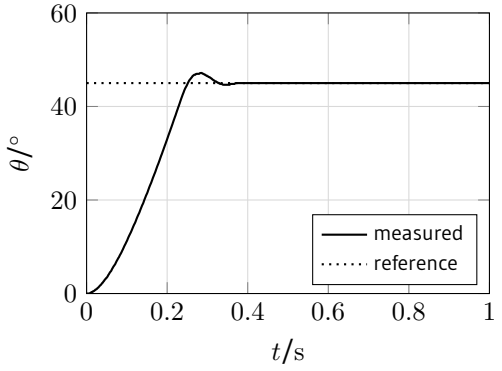


Figure C.6: Measured step response of run 6 (controller gains see Table 3.1)

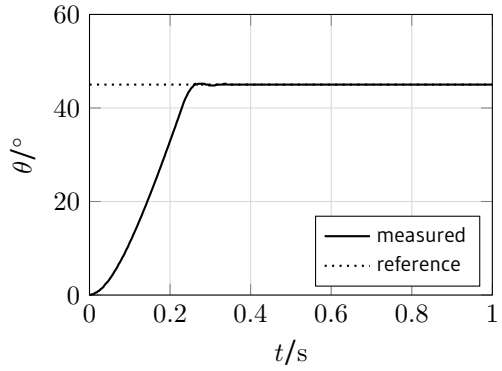


Figure C.7: Measured step response of run 7 (controller gains see Table 3.1)

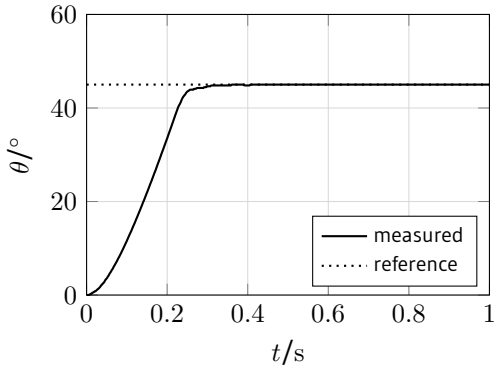


Figure C.8: Measured step response of run 8 (controller gains see Table 3.1)

Chapter D

Disturbance and Sinusoidal Reference

Figure D.1 shows the reaction to a disturbance applied manually to the pointer with the controller gains of run 1. For the tracking of a sinusoidal reference with an amplitude of $10°$ and an angular frequency of $5\,\mathrm{rad\,s^{-1}}$, K_P is varied while K_I and K_D are kept constant. The observations are listed in Table D.1; the tracking is shown in Figure D.2, and the control error and control value for each gain in Figures D.3 to D.7.

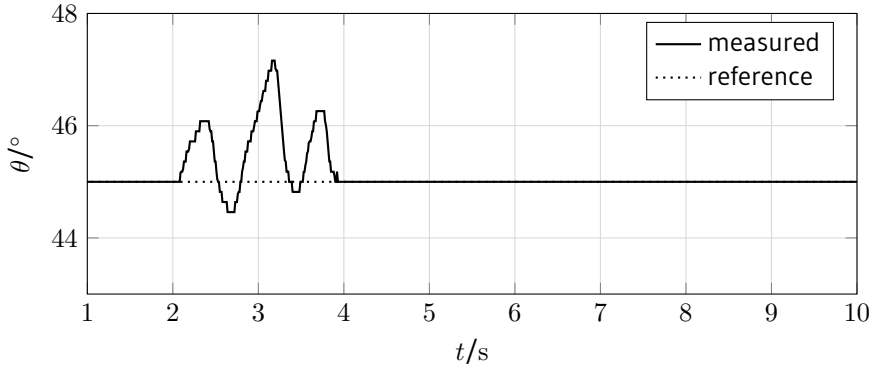


Figure D.1: Measured angular position after the reference step of $45°$ with a disturbance applied manually to the pointer (controller gains of run 1, section from $1\,\mathrm{s}$ to $10\,\mathrm{s}$)

Table D.1: Proportional gains and observations for the tracking of a sinusoidal reference (amplitude $10°$, angular frequency $5\,\mathrm{rad\,s^{-1}}$) with $K_\mathrm{I} = 22.2\,\mathrm{V\,rad^{-1}\,s^{-1}}$ and $K_\mathrm{D} = 4.28\,\mathrm{V\,s\,rad^{-1}}$

$K_\mathrm{P}/(\mathrm{V\,rad^{-1}})$	Observation
195	slightly slow
300	control error halved
600	control error further reduced
800	control value too noisy
1200	control value strongly noisy

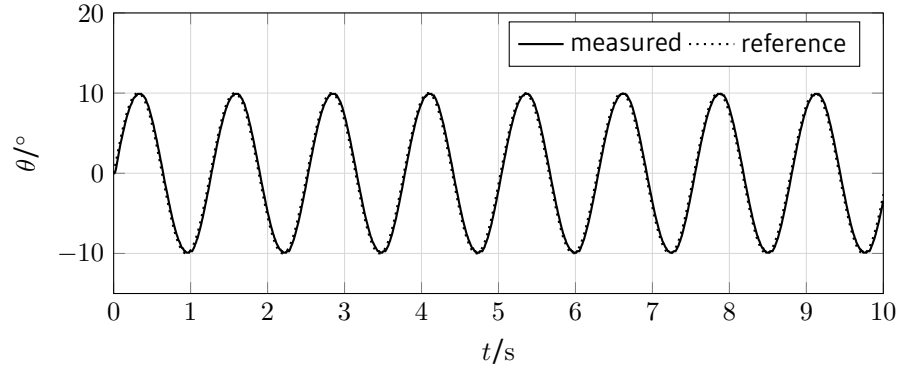


Figure D.2: Measured angular position and sinusoidal reference for $K_\mathrm{P} = 195\,\mathrm{V\,rad^{-1}}$

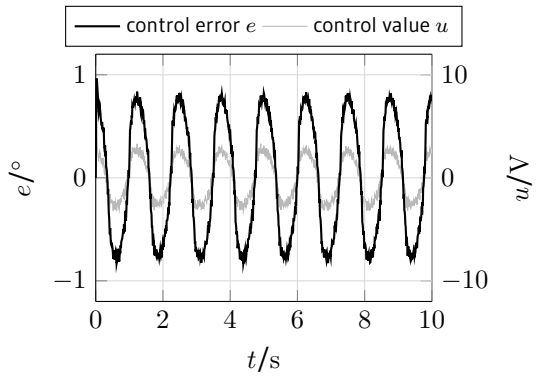


Figure D.3: Sinusoidal reference: control error and control value for $K_\mathrm{P} = 195\,\mathrm{V\,rad^{-1}}$

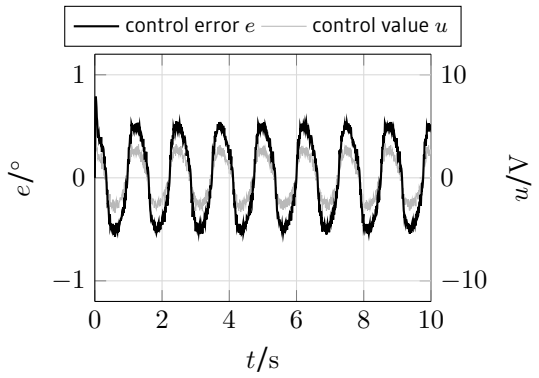


Figure D.4: Sinusoidal reference: control error and control value for $K_\mathrm{P} = 300\,\mathrm{V\,rad^{-1}}$

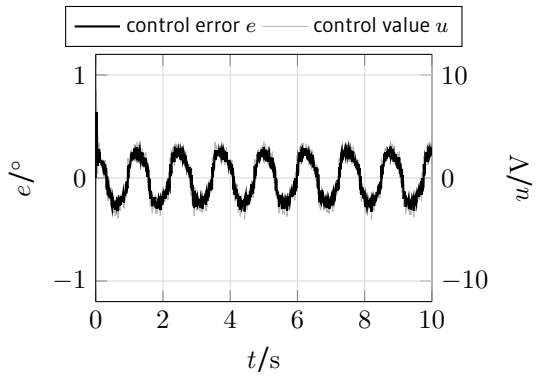


Figure D.5: Sinusoidal reference: control error and control value for $K_\mathrm{P} = 600\,\mathrm{V\,rad^{-1}}$

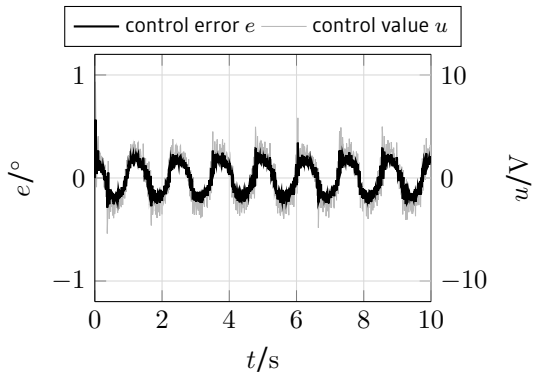


Figure D.6: Sinusoidal reference: control error and control value for $K_\mathrm{P} = 800\,\mathrm{V\,rad^{-1}}$

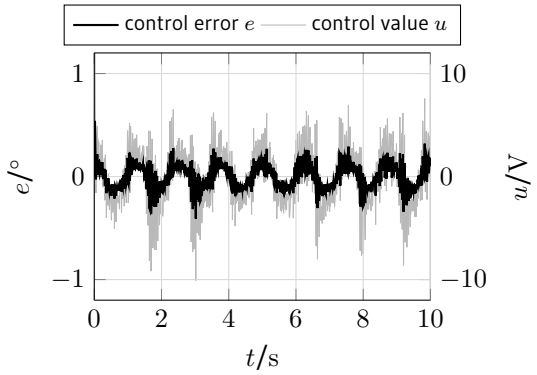


Figure D.7: Sinusoidal reference: control error and control value for $K_\mathrm{P} = 1200\,\mathrm{V\,rad^{-1}}$